Requirements Elicitation

When eliciting requirements, we fundamentally based our ideas on the given product brief and the future stakeholder meeting. This resulted in a system where we first summarised the requirements into one Single Statement of Need (SSON), then broke them down into the given User and System requirements, then broke those categories down into topics, then each individual requirement - which would then become questions for the stakeholder. This decomposition allowed us to simplify and ensure we secured the fundamental requirements from the stakeholder. It also allowed us to frame the requirements we brought to the user in the context of the product brief and it allowed us to more easily maintain focus on the basics of what was needed when preparing for the meeting.
This follows through in the way we would present our presumed requirements to the stakeholder. By sticking to each topic it can both make it easier to manage personally as well as guiding future development as it provides clear lines between the systems we develop and what the user expects from said systems.

Single Statement Of Need (SSON)

As above, we first began with a SSON. We chose to do this because it gave us a clear, simple, guiding concept to work our requirements around. By using one sentence rather than a whole document it provides one clear goal. This enables us to work around a framework that always centers the stakeholder and their given product brief when developing
The SSON we decided on was "*A single-player game where a player controls an avatar, moving around the university campus and attempts to escape from the geese*". We felt this provided a concise but accurate depiction of the product brief whilst maintaining the ideal short and simple nature of an SSON.

System and User Requirements

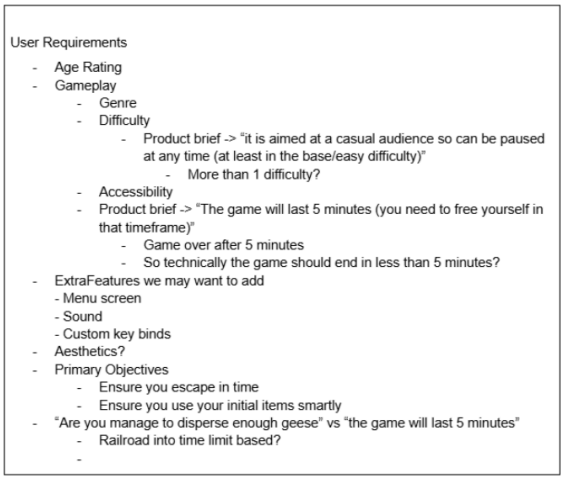
We then broke our theoretical requirements into User and System requirements. This allows us to separate the creative and practical requirements in a way that will benefit development. This is because by keeping a set of System requirements, which need to run throughout the entire system, we can focus on how the system **must** work throughout each individual User requirement.

Topics

We then split our User requirements into topics. This adds a further level of decomposition which makes the project overall easier to manage and prevent further overlap in work as everyone on the project knows the requirements they need fall under a certain requirement found in the product brief.

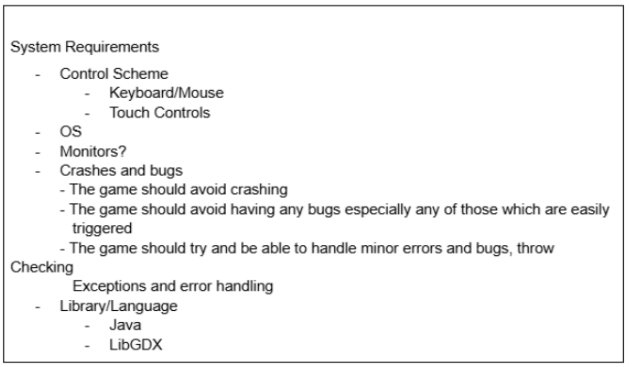
Planning

Below is our initial planning of user requirements. These were made in a meeting from which we would take them. We would then simplify further, or expand certain ideas based on the product brief



User Requirements

- Age Rating
- Gameplay
  - Genre
  - Difficulty
    - Product brief -> "it is aimed at a casual audience so can be paused at any time (at least in the base/easy difficulty)"
      - More than 1 difficulty?
  - Accessibility
  - Product brief -> "The game will last 5 minutes (you need to free yourself in that timeframe)"
    - Game over after 5 minutes
    - So technically the game should end in less than 5 minutes?
- ExtraFeatures we may want to add
  - Menu screen
  - Sound
  - Custom key binds
- Aesthetics?
- Primary Objectives
  - Ensure you escape in time
  - Ensure you use your initial items smartly
- "Are you manage to disperse enough geese" vs "the game will last 5 minutes"
  - Railroad into time limit based?
  -



System Requirements

- Control Scheme
  - Keyboard/Mouse
  - Touch Controls
- OS
- Monitors?
- Crashes and bugs
  - The game should avoid crashing
  - The game should avoid having any bugs especially any of those which are easily triggered
  - The game should try and be able to handle minor errors and bugs, throw Checking
    Exceptions and error handling
- Library/Language
  - Java
  - LibGDX

From this basis, we produced a document of our full list of requirements and questions for the user: Interview plan for before the interview, Interview Notes for during the interview.
We took this interview plan with us into the stakeholder meeting. By linking these inherently we made sure we maintained focus on the stakeholder through the entire process of analysing and eliciting based off of the product brief.

Requirements Negotiation

After finishing our requirements meeting, we produced a list of final requirements based off of both the answers to our questions and our given topics system. Most requirements we discussed were accepted so few differences between the elicited and negotiated requirements exist.

Single Statement Of Need

- This is what we interpreted the project as overall from your brief, you want:
  A single-player game where a player controls an avatar, moving around the university campus and attempts to escape from the geese.

Functional requirements (FR) / User Requirements

ID	Description
Player Movement and Control	
FR_MOV_01	The game shall provide standard 2-dimensional (X / Y axis) movement for the player in a top-down perspective.
FR_MOV_02	Player movement and actions shall be controlled using a combination of keyboard and mouse inputs.
FR_ATK_01	The player shall project "bread crumbs" to distract and disperse enemy entities upon contact.

Enemies AI and Behaviour	
FR_ENA_01	Enemies shall be represented as geese entities within the game environment.
FR_ENA_02	Enemy entities shall dynamically seek and follow the player's position in real-time.
FR_ENA_03	The engine shall render and update a large density of simultaneous enemy entities without degrading game's performance.
FR_ENA_04	Enemy entities shall not feature blood or graphic violence. Upon colliding with player projectiles, enemy entities shall play a "dispersal" visual sequence and be removed from the active play area.
Items	
FR_ITM_01	The game map shall spawn floor items that grant temporary combat advantages or additional projectile capabilities to the player.
FR_ITM_02	All in-game collectibles and consumables shall be visually styled as food items.
Events	
FR_EVT_01	The map shall feature static landmark objects that trigger randomised game events when the player enters their defined proximity radius.
FR_EVT_02	Dynamic events shall apply distinct global or localised modifications to active gameplay mechanics (e.g. speed modifiers, altered spawn rates).
FR_EVT_03	Event triggers and active event statuses shall provide clear, unmistakable visual cues to the user.
Session State and Score	
FR_SSS_01	Dispersing enemy entities shall increment the player's score. Collisions with enemy entities shall decrement the score by a fixed penalty amount.
FR_SSS_01	The system shall grant a victory state immediately upon the player reaching a predefined target score threshold.
FR_SSS_02	The game session shall end in a failure state if: ● The global session timer reaches zero (5-minute limit). ● The player's total score drops to zero.
User Interface and Navigation	
FR_GUI_01	The system shall load a dedicated Main Menu screen upon initialisation, prior to entering active gameplay.
FR_GUI_02	The UI shall display clear instructions covering controls, core

	mechanics, and primary win/lose objectives.
FR_GUI_03	The system shall include an in-game Pause Menu distinct from the Main Menu, accessible during active gameplay via a dedicated key input.

Non-Functional Requirements (NFR) / System Requirements

ID	Description
Architecture and Portability	
NFR_SYS_01	The software shall be written and compiled targeting Java 17.
NFR_SYS_02	The application shall run natively on Windows and Linux distributions, maintaining full functional and visual parity across both operating systems.
NFR_SYS_03	The application layout shall dynamically scale to support display sizes ranging from 13-inch laptop screens up to 27-inch desktop monitors.
Performance and Constraints	
NFR_PRF_01	The game shall be optimised for baseline lab hardware specifications (Intel Core i5-14500, NVIDIA RTX A1000 GPU, 16GB RAM).
NFR_PRF_02	Execution time from launch to the Main Menu shall not exceed 10 seconds on target hardware.
NFR_PRF_03	A single gameplay session shall be strictly hard-capped at a maximum run time of 5 minutes (300 seconds).